

AWS CloudTrail Monitoring for EC2 Activity using Wazuh SIEM

Project Description

This project focuses on monitoring **AWS EC2 instance activities** using **AWS CloudTrail** and integrating the logs into **Wazuh SIEM** for centralized security visibility.

By collecting CloudTrail logs in Amazon S3 and ingesting them into Wazuh, the SOC analyst can track **instance start, stop, terminate, and other management actions** in real time.

Project Goal

To gain **visibility and control over EC2 management activities** by detecting and monitoring all AWS API actions related to EC2 instances through Wazuh.

Project Objectives

- Enable AWS CloudTrail for EC2 management events
- Securely store logs in an S3 bucket
- Allow EC2 instances to access CloudTrail logs via IAM roles
- Ingest CloudTrail logs into Wazuh Manager
- Visualize EC2 activity events in Wazuh Dashboard

Why This Project is Important

- EC2 instances are **critical cloud assets**
- Unauthorized start/stop or termination can cause **security incidents**
- CloudTrail provides **audit-level visibility**
- Wazuh helps reduce blind spots by **centralized monitoring**
- Essential skill for **SOC, Cloud Security & SIEM roles**

Architecture Overview

AWS CloudTrail → S3 Bucket → EC2 IAM Role → Wazuh Manager → Wazuh Dashboard

Implementation Steps

Step 1: AWS CloudTrail Creation & Configuration

AWS CloudTrail was created as a **multi-region trail** to capture EC2-related management events across the AWS account.

During creation, management events were enabled for both **read and write API calls**, ensuring EC2 start, stop, and terminate actions are logged.

Choose trail attributes

General details

A trail created in the console is a multi-region trail. [Learn more](#)

Trail name

Enter a display name for your trail.

waz-aws-cloudtrails

3-128 characters. Only letters, numbers, periods, underscores, and dashes are allowed.

☐ Enable for all accounts in my organization

To review accounts in your organization, open AWS Organizations. [See all accounts](#)

Storage location [Info](#)

☒ Create new S3 bucket

Create a bucket to store logs for the trail.

☐ Use existing S3 bucket

Choose an existing bucket to store logs for this trail.

Trail log bucket and folder

Enter a new S3 bucket name and folder (prefix) to store your logs. Bucket names must be globally unique.

aws-cloudtrail-logs-503561425397-1539985b

Logs will be stored in aws-cloudtrail-logs-503561425397-1539985b/AWSLogs/503561425397

Log file SSE-KMS encryption [Info](#)

☒ Enabled

Customer managed AWS KMS key

☒ New

☐ Existing

AWS KMS alias

waz-alias

KMS key and S3 bucket must be in the same region.

▼ Additional settings

Log file validation [Info](#)

☒ Enabled

SNS notification delivery [Info](#)

☐ Enabled

Step 2: Log Storage & Security Configuration

An **Amazon S3 bucket** was configured during CloudTrail setup to store audit logs securely. Log file validation and **SSE-KMS encryption** were enabled to maintain log integrity and prevent unauthorized tampering.

Choose log events

Events [Info](#)

Record API activity for individual resources, or for all current and future resources in AWS account. [Additional charges apply](#)

Event type

Choose the type of events that you want to log.

☒ Management events

Capture management operations performed on your AWS resources.

☐ Data events

Log the resource operations performed on or within a resource.

☐ Insights events

Identify unusual activity, errors, or user behavior in your account.

☐ Network activity events

Network activity events provide information about resource operations performed on a resource within a virtual private cloud endpoint.

Management events [Info](#)

Management events show information about management operations performed on resources in your AWS account.

ⓘ Multiple management events trails detected. Charges apply to duplicated logged management events. [Additional charges apply](#)

API activity

Choose the activities you want to log.

☒ Read

☒ Write

☐ Exclude AWS KMS events

☐ Exclude Amazon RDS Data API events

Cancel

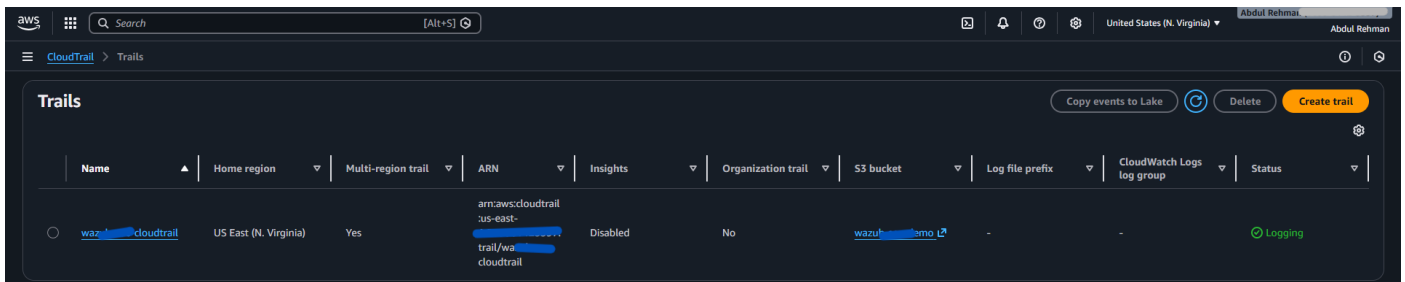
Previous

Next

Step 3: CloudTrail Status Verification

After configuration, the CloudTrail was verified in the AWS console to confirm that **logging is enabled** and actively delivering logs to the S3 bucket.

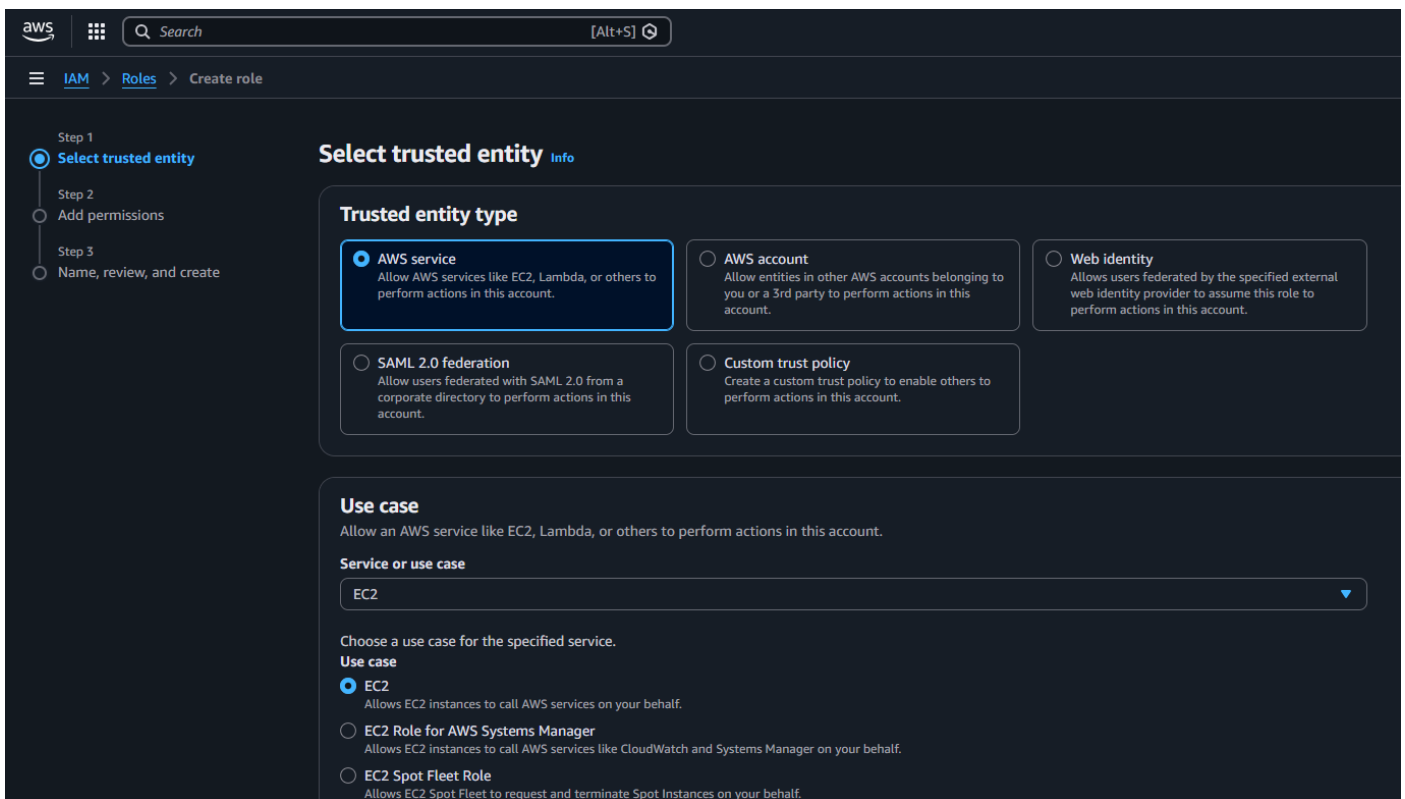
This confirms successful trail creation and readiness for SIEM ingestion.



Step 4: Creating IAM Role for EC2

An IAM role was created with **trusted entity: EC2**.

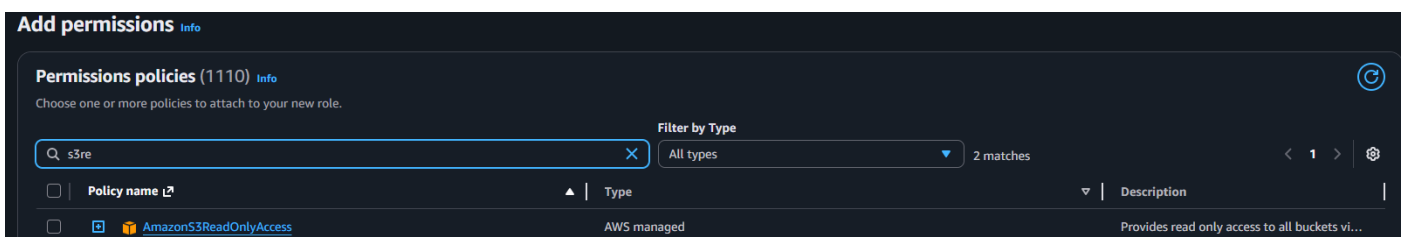
This role allows the EC2 instance to securely access CloudTrail logs without using hardcoded credentials.



Step 5: Assigning S3 Read Permissions

The IAM role was attached with **AmazonS3ReadOnlyAccess** policy.

This enables the EC2 instance (Wazuh agent) to read CloudTrail logs from the S3 bucket.



Step 6: Configuring Wazuh CloudTrail Module

The CloudTrail **wodle block** was added to the `ossec.conf` file on the Wazuh Manager.

The configuration was set to run every minute and read logs directly from the CloudTrail S3 bucket.

```
<!-- ===== -->
<!-- AWS CLOUDTRAIL (S3) -->
<!-- ===== -->
<wodle name="aws-s3">
  <disabled>no</disabled>

  <!-- Run every 1 minute -->
  <interval>1m</interval>
  <run_on_start>yes</run_on_start>

  <!-- Do not fail manager on errors -->
  <skip_on_error>yes</skip_on_error>

  <!-- CloudTrail bucket -->
  <bucket type="cloudtrail">
    <name>wa-...-demo</name>
  </bucket>
</wodle>
```

Step 7: Restarting Wazuh Manager

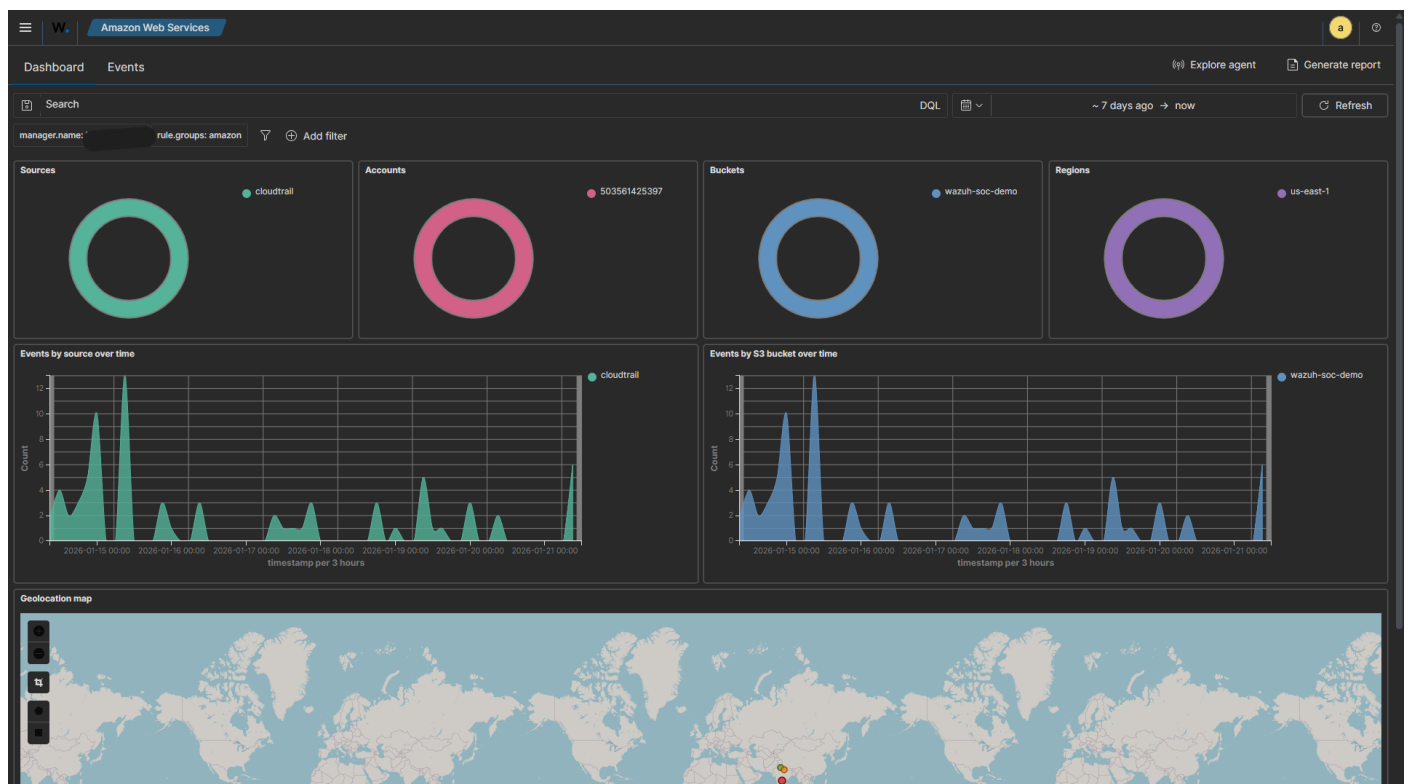
After configuration changes, the Wazuh Manager service was restarted.

This ensured the new CloudTrail integration was loaded successfully.

Step 8: Monitoring EC2 Events in Wazuh Dashboard

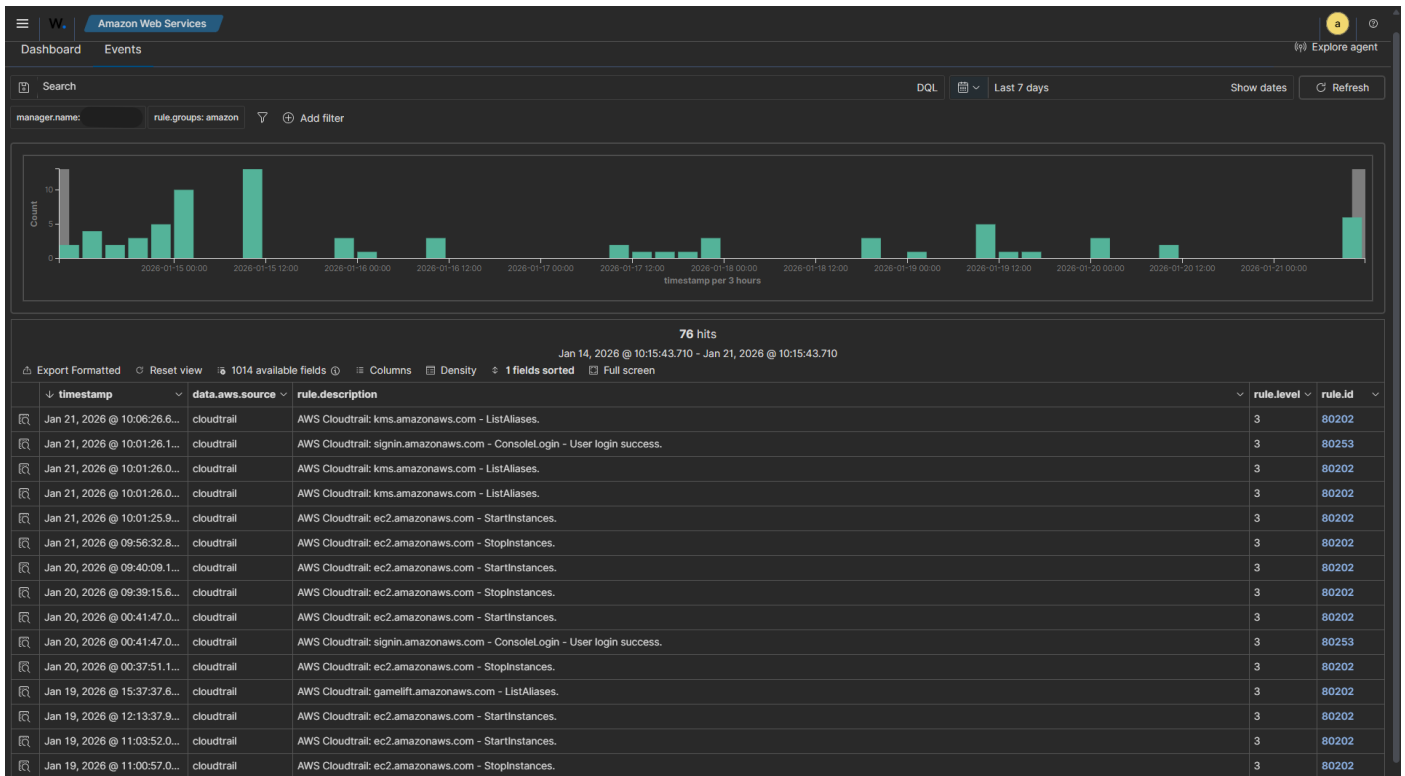
After enabling the CloudTrail integration, AWS audit logs started appearing in the **Wazuh Dashboard**.

EC2-related management events were successfully ingested and displayed with timelines and summary visualizations.



Step 9: Visualizing EC2 Activity Events

Individual CloudTrail events such as **StartInstances** and **StopInstances** were visible in the Events view. Each event includes timestamp, AWS service, action performed, and rule details, enabling effective EC2 activity analysis.



Project Benefits

- Centralized AWS activity monitoring
- Improved cloud security visibility
- Faster detection of unauthorized actions
- SOC-ready logging and investigation
- Reduced cloud misconfiguration risks

Skills Gained

- AWS CloudTrail configuration
- IAM role and policy management
- S3 security and log storage
- Wazuh SIEM integration
- Cloud security monitoring
- SOC investigation workflows

Conclusion

This project demonstrates how **AWS CloudTrail** and **Wazuh SIEM** can be effectively integrated to monitor EC2 activity in a cloud environment.

It showcases practical SOC skills in **cloud auditing, detection, and log analysis**, making it highly relevant for real-world security operations.